

EchoVision

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ABSTRACT

What is it like to be a bat? "EchoVision" is a mixed reality art experience that simulates bat echolocation, letting participants step into a bat's perceptual world. Using a custom-designed, handheld, bat-shaped mask built with the open-source HoloKit headset, participants explore echolocation through visual feedback that responds to their vocalizations as they bounce off the surrounding environment—all detected by LiDAR Scanner. Through this artistic simulation, users discover how bats navigate and perceive their world, fostering empathy for non-human perspectives.

Index Terms: Augmented Reality, Mixed Reality, Handheld Mixed Reality Mask, Ecocentric design, Eco-Phenomenology

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1 PROJECT URL

<https://reality.design/project/echovision>

2 DESCRIPTION OF ARTWORK

2.1 Thematic Statement

Bats have long been misunderstood creatures, victims of misconceptions and negative stereotypes. Common misconceptions portray them as inherently dangerous or primary disease carriers, unfortunately exacerbated by misinformation during the COVID-19 pandemic, despite the lack of conclusive evidence linking bats directly to the origins of the COVID-19 virus. Contrary to these unfounded myths, bats play vital ecological roles: they control pests, pollinate plants, and disperse seeds, supporting various ecosystems that are essential for maintaining ecological balance and biodiversity. The gap between public perception and ecological reality has led to the persecution of bat populations. To protect these vital creatures and their habitats, widespread education and awareness are urgently needed.

The tension between misconceptions about bats and their ecological significance raises a profound question: Can we truly understand bats? Philosopher Thomas Nagel's seminal 1974 essay "What Is It Like to Be a Bat?" explores this inquiry, using bats as a prime example to delve into the subjective nature of consciousness. He argued that the unique sensory world of bats, particularly their

use of echolocation, presents an insurmountable barrier to human understanding. He contended that our capacity to comprehend the subjective experiences of other creatures is inherently constrained by the fundamental differences between our perception and those of other species. Although we can analyze and explain the scientific principles of bat echolocation, we cannot fully grasp how a bat perceives and navigates its surroundings through this sensory system. This philosophical inquiry challenges us to consider the limits of our empathy and comprehension of non-human experiences.

Responding to the IEEE VR 2025 XR Gallery theme "Questioning the truthfulness of our digital reality," which aspires to transcend harsh realities of conflict towards a wave of unity, we pose a crucial question: Can we foster a more-than-human, ecocentric perspective by offering the public an embodied experience of bats' perceptual world through cutting-edge media technologies, rather than relying on abstract knowledge? We aim to influence society by sparking thought-provoking ecological awareness and scientific education.

2.2 Artwork

Inspired by the eco-phenomenological approach, which transcends abstract knowledge to foster a more intimate, sensory-rich embodied experience, we have created EchoVision—a Mixed Reality (MR) art experience offering a momentary glimpse into the perceptual world of bats. Using a handheld, bat-shaped mask encasing the open-source HoloKit MR headset, this experience immerses participants by simulating bat echolocation through sound visualization. As users vocalize, the system visualizes sound echoes superimposed on the real world, propagating through real-time 3D reconstructed contours of the environment—created by the LiDAR scanner of the smartphone attached to the mask. EchoVision leverages the parallels between bat echolocation and LiDAR technology: both emit signals (laser pulses for LiDAR, ultrasonic waves for bats) and use the reflected signals to create detailed environmental maps. This artistic simulation allows users to understand how bats navigate and perceive their surroundings. By offering this unique glimpse into a bat's perceptive world, EchoVision not only fosters an empathetic understanding of non-human sensory experiences but also emphasizes the profound differences between human and bat perception.

3 ARTIST INFORMATION

Botao 'Amber' Hu is a social computing researcher and experiential futures designer. He directs Reality Design Lab (<https://reality.design>), an independent interdisciplinary research and design lab exploring the intersection of soma design (bodily play), speculative design (artificial life), spatial computing (mixed reality), and social computing (protocol science, public-interest cryptography, decentralized AI governance). His primary focus is designing socially experiential futures using mixed reality as the main medium. He also serves as a visiting lecturer at the China Academy of Art. His works have been featured at top conferences such as SIGGRAPH, CSCW, CHI, UbiComp, WWW, TEI, ISEA, IEEEVR, IEEEVIS, ISMAR, Ars Electronica, SXSW, and TEDx and have received accolades including the SIGGRAPH Best in Show, CHI Best Interactivity, Webby, Red Dot, iF Design, Good

Design, A' Design, Core77 Design Award, and grants from the Ethereum Foundation. He holds a bachelor's degree in computer science from Tsinghua University and a master's degree in computer science with AI concentration from Stanford University.

Jiabao Li is a designer, artist, creative technologist, Assistant Professor at The University of Texas at Austin, Founding Director of Ecocentric Future Lab, Co-founder and Chief Design Officer of Endless Health. She creates works addressing planetary and human health, humane technology, and perceptions. In Jiabao's TED Talk, she uncovered how technology mediates the way we perceive reality. At Apple, she invented and explored new technologies for future products like the Apple Vision Pro. Jiabao has received numerous awards including Forbes China 30 Under 30, the Outstanding Professor Award at MoMA, the STARTS Prize, and Fast Company. Her work has been exhibited internationally, at MoMA, Venice Architecture Biennale, Ars Electronica, Exploratorium, Future of Today Biennial, DesignArt Tokyo, Milan, and Dubai Design Week. Her academic papers have been published in top conferences including SIGGRAPH, CHI, IEEE VIS, and Nature sub-journals. She graduated from Harvard GSD with Distinction and thesis award.

Danlin Huang is an undergraduate student at the China Academy of Art, majoring in Industrial Design. Her media artworks include "Body Oracle" and "Foldiverse." As an artist, speculative design researcher, and AR developer, her academic papers have been published in top conferences such as SIGGRAPH Asia and CHI PLAY. Her works have been exhibited at the China Digital Art Exhibition and the China National Art Exhibition, among others. Huang's work has earned her multiple accolades, including the Platinum Creative Award, the Gold Award at the Rokid University AR Innovation Challenge, the NCDA Gold Award, and the CADA Gold Award.

Jianan Johanna Liu is an undergraduate student at the China Academy of Art, majoring in Art and Technology. She leads and programs the AR application "Foldiverse" while serving as a teaching assistant for the "Speculative Spatial Computing Design" course. Liu's work as an AR developer and accessibility researcher has led to publications in prestigious conferences, including SIGGRAPH Asia, CHI PLAY, and CSCW.

Xiaobo Aaron Hu is a creative technologist, VJ, and psychological researcher. His work focuses on the intersection of technology and psychology, developing new methods for psychological measurement and intervention. He also investigates how technological advancements impact societal psychology and moral ethics. Aaron's background includes leading the innovation lab at 4A Company, with over a decade of experience in new media, experiential marketing, and digital innovation. Skilled in both software and hardware, he excels at delivering optimal technological solutions.

Yilan Elan Tao is a designer and researcher specializing in HCI and Mixed Reality. Tao's exceptional design and presentation skills have been crucial to the success of Reality Design Lab's award-winning projects HoloKit, MOFA, GravField, etc. These projects have been showcased at prestigious conferences and venues such as CHI, SIGGRAPH, ISMAR, iF award, Red Dot Award, Core77 Awards, and SXSW. She holds a Master's degree in Architecture from Cornell University.